REPLACES CRON-ONLY POLLING

CRM Sync — Event-Driven Integration Spec

Version 1.0 · 2026-05-27 · Specification. Cron is the wrong primitive for data integration — most polls find nothing and the hits arrive up to fifteen minutes stale. This spec moves the external integrations to an event-driven substrate: **data moves when something happens, not when a clock ticks**, with ordering, replay, and observability defined per channel, and cron demoted to the reconcile backstop.

Version: **1.0** Date: **2026-05-27** Status: **Specification**

Replaces: **Cron-only polling for external integrations**

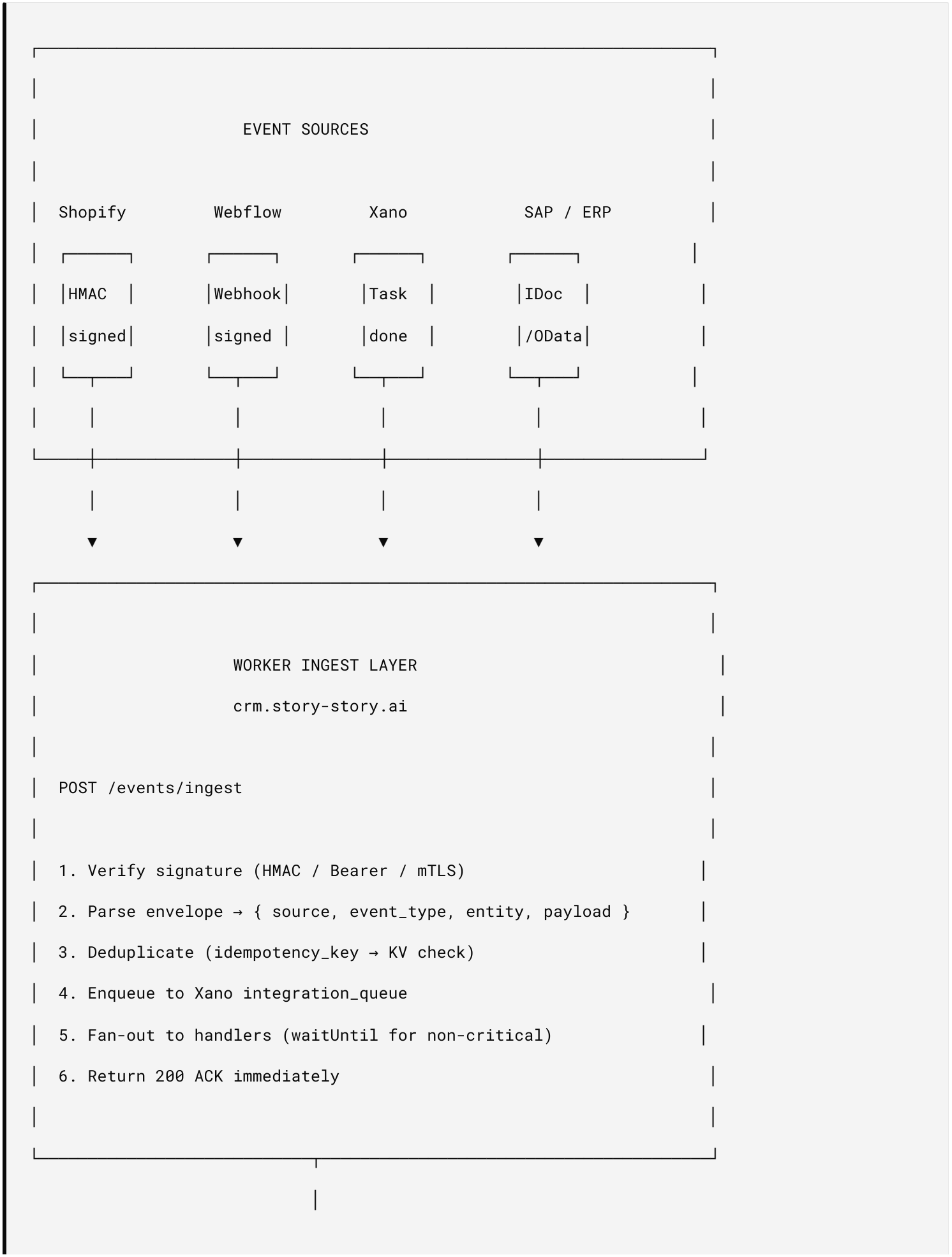
1. Problem with Cron Polling

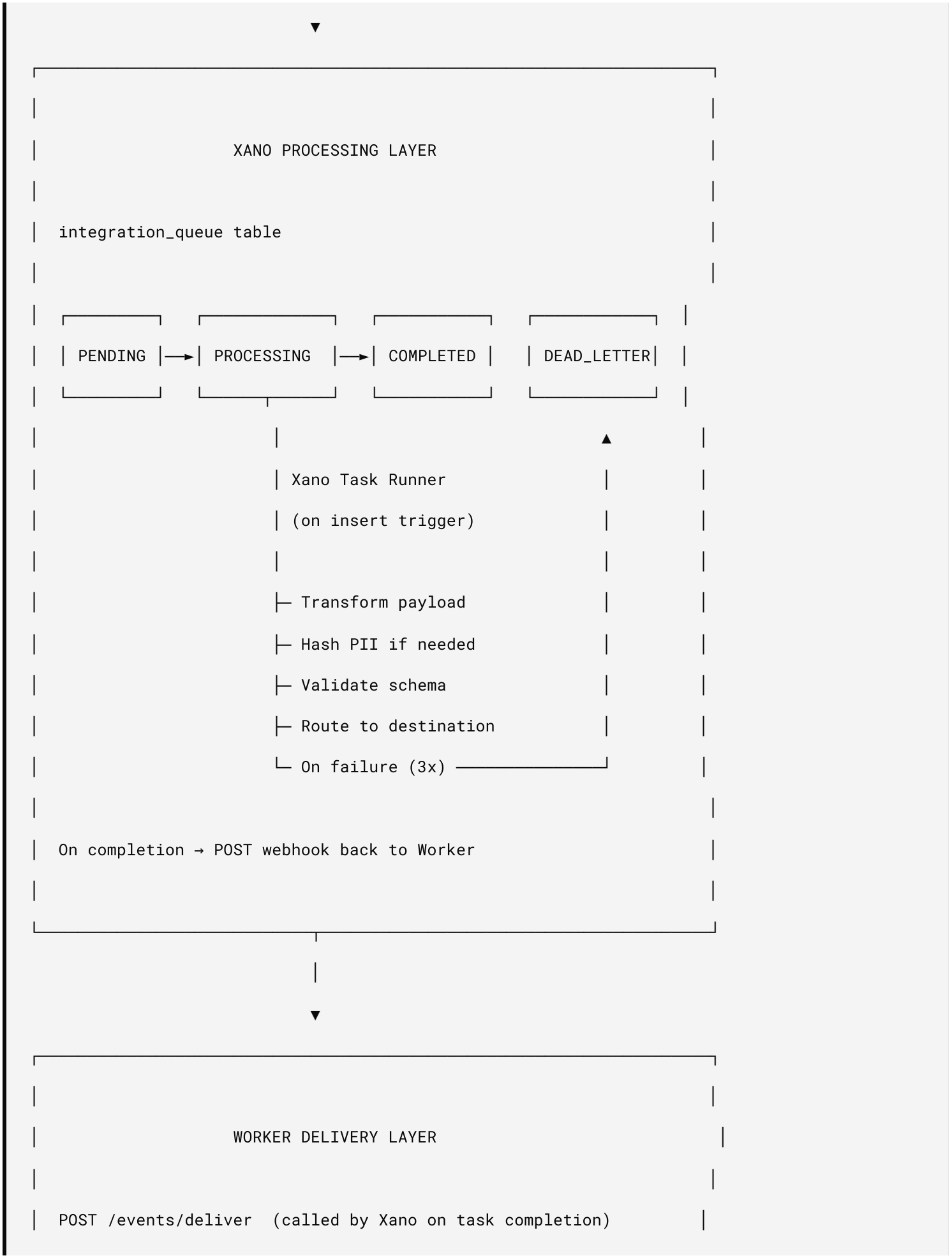
```
Cron (*/15 * * * *)
  |
  ├─ 95% of runs: no new data → wasted compute
  ├─ 5% of runs: data ready → up to 15 min stale
  ├─ No ordering guarantee across runs
  └─ Scales poorly: more channels = more polls per tick
```

Cron is the wrong primitive for data integration. It was designed for periodic maintenance (token refresh, customer sync), not for reactive data flows.

2. Event-Driven Architecture

Principle: Data moves when something happens, not when a clock ticks.


EVENT SOURCES
Shopify
Webflow
Xano
SAP / ERP
HMAC signed
Webhook signed
Task done
IDoc /OData
WORKER INGEST LAYER
crm.story-story.ai
POST /events/ingest
1. Verify signature (HMAC / Bearer / mTLS)
2. Parse envelope → { source, event_type, entity, payload }
3. Deduplicate (idempotency_key → KV check)
4. Enqueue to Xano integration_queue
5. Fan-out to handlers (waitUntil for non-critical)
6. Return 200 ACK immediately


XANO PROCESSING LAYER
integration_queue table
PENDING
PROCESSING
COMPLETED
DEAD_LETTER
Xano Task Runner
(on insert trigger)
Transform payload
Hash PII if needed
Validate schema
Route to destination
On failure (3x)
On completion → POST webhook back to Worker
WORKER DELIVERY LAYER
POST /events/deliver (called by Xano on task completion)

```
│                                                                  │
│  Routes by channel:                                              │
│  ├─ sap       → OData POST / IDoc SFTP                           │
│  ├─ nielseniq → S3 drop / SFTP                                   │
│  ├─ circana   → API POST                                         │
│  ├─ ga4       → Measurement Protocol                             │
│  ├─ adobe     → AEP Streaming Ingestion                          │
│  ├─ webflow   → CMS API                                          │
│  └─ r2        → Archive (always, parallel to primary delivery)   │
│                                                                  │
│  On success → PATCH /events/{id}/status = delivered              │
│  On failure → PATCH /events/{id}/status = failed + retry logic   │
│                                                                  │
└──────────────────────────────────────────────────────────────────┘
```

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3. Event Envelope Schema

Every event entering the system uses a standard envelope, regardless of source:

```
interface EventEnvelope {
  // Identity
  event_id: string;              // UUID — set by source or generated on ingest
  idempotency_key: string;       // SHA-256(source + entity_id + event_type + timestamp)

  // Routing
  source: EventSource;           // "shopify" | "webflow" | "xano" | "sap" | "erp" | "wms" | "m
  event_type: string;            // "customer.updated" | "order.created" | "inventory.changed"
  entity_type: EntityType;       // "customer" | "order" | "product" | "inventory" | "invoice"
  entity_id: string;             // Source-system ID

  // Scoping
  tenant_shop: string;           // Multi-tenant isolation

  // Payload
  payload: Record<string, unknown>;  // Source-specific data

  // Metadata
  timestamp: string;             // ISO 8601 — when the event occurred (not when received)
  received_at: string;           // When worker ingested it

  // Delivery
  destinations: string[];        // ["sap", "ga4", "r2"] — fan-out targets
  priority: 1 | 2 | 3 | 4 | 5;  // 1=critical (inventory), 5=low (analytics batch)
}
```

4. Worker: Event Ingest Endpoint

`POST /events/ingest`

Single entry point for all external events. Replaces per-source webhook endpoints over time.

```
async function handleEventIngest(
  cfg: ResolvedConfig,
  env: Env,
  request: Request,
  origin: string | null,
  rotatableKey: string | null
): Promise<Response> {

  // 1. Authenticate — support multiple auth methods
  const authResult = await authenticateEventSource(cfg, env, request, rotatableKey);
  if (!authResult.authenticated) {
    return jsonResponse({ error: "Unauthorized", method: authResult.method }, 401, origin);
  }

  // 2. Parse envelope
  const raw = await request.json() as Partial<EventEnvelope>;
  const envelope = normalizeEnvelope(raw, authResult.source);

  // 3. Idempotency check — reject duplicates
  const dedupeKey = `evt:${envelope.idempotency_key}`;
  const existing = await env.CRM_STATE.get(dedupeKey);
  if (existing) {
    return jsonResponse({ ok: true, deduplicated: true, event_id: envelope.event_id }, 200, ori
  }

  // 4. Store idempotency marker (TTL: 7 days)
  await env.CRM_STATE.put(dedupeKey, envelope.event_id, { expirationTtl: 604800 });

  // 5. Enqueue to Xano for processing
```

```
  const queued = await enqueueToXano(cfg, envelope);

  // 6. Fire-and-forget side effects (non-blocking)
  //    These run after the 200 response via waitUntil pattern
  const ctx = { waitUntil: (p: Promise<unknown>) => p }; // simplified

  // Immediate fan-out for low-latency destinations
  if (envelope.destinations.includes("ga4") && envelope.priority <= 2) {
    // GA4 gets real-time push for high-priority events
    ctx.waitUntil(pushEventToGA4(cfg, envelope));
  }
  if (envelope.destinations.includes("r2")) {
    // Always archive
    ctx.waitUntil(archiveToR2(env, envelope));
  }

  // 7. ACK immediately — Xano handles the rest
  return jsonResponse({
    ok: true,
    event_id: envelope.event_id,
    queued: queued,
    destinations: envelope.destinations,
  }, 202, origin);  // 202 Accepted — processing async
}
```

Authentication per source

```
async function authenticateEventSource(
  cfg: ResolvedConfig,
  env: Env,
  request: Request,
  rotatableKey: string | null
): Promise<{ authenticated: boolean; source: string; method: string }> {

  const url = new URL(request.url);
  const sourceHint = url.searchParams.get("source") || request.headers.get("x-event-source") ||

  // Shopify HMAC
  if (sourceHint === "shopify" || request.headers.has("x-shopify-hmac-sha256")) {
    const { valid } = await verifyShopifyHmac(request, cfg.shopifyAppSecret);
    return { authenticated: valid, source: "shopify", method: "hmac" };
  }

  // SAP client certificate or OAuth token
  if (sourceHint === "sap") {
    const token = request.headers.get("authorization")?.replace("Bearer ", "");
    if (token) {
      const valid = await verifySAPToken(cfg, token);
      return { authenticated: valid, source: "sap", method: "oauth2" };
    }
  }

  // Xano task callback (shared secret)
  if (sourceHint === "xano" || request.headers.has("x-xano-task-id")) {
    const secret = request.headers.get("x-xano-webhook-secret") || "";
```

```
    const valid = secret === cfg.xanoWebhookSecret;
    return { authenticated: valid, source: "xano", method: "shared_secret" };
  }

  // Admin key / rotatable key (manual, ERP generic, testing)
  if (verifyBearerToken(cfg as unknown as Env, request, undefined, rotatableKey)) {
    return { authenticated: true, source: sourceHint || "admin", method: "admin_key" };
  }

  return { authenticated: false, source: "unknown", method: "none" };
}
```

---

5. Xano: Task Runner (Not Cron)

How Xano Task Runner works

Xano's task system is **event-triggered**, not time-based:

```
INSERT into integration_queue (status = "pending")
         │
         ▼
  Xano DB Trigger (on insert)
         │
         ▼
  Xano Background Task starts
         │
         ├─ Read message from queue
         ├─ Transform payload per destination schema
         ├─ Hash PII where required (consent-gated)
         ├─ Validate output schema
         ├─ Update status → "ready"
         ├─ POST webhook → Worker /events/deliver
         │
         ▼
  Task completes — no polling, no cron
```

Xano API Endpoint: `POST /api:{group}/integration-enqueue`

Called by Worker to insert events into the queue:

```
POST /api:{group}/integration-enqueue

Authorization: Bearer {XANO_API_KEY}

Content-Type: application/json


{

  "event_id": "evt_a1b2c3d4",

  "source": "shopify",

  "event_type": "customer.updated",

  "entity_type": "customer",

  "entity_id": "gid://shopify/Customer/12345",

  "tenant_shop": "hx-stage.myshopify.com",

  "payload": { ... },

  "destinations": ["sap", "ga4", "r2"],

  "priority": 2,

  "timestamp": "2026-05-27T14:30:00Z"

}


Response: { "ok": true, "queue_id": 4521 }
```

Xano DB Trigger: `on_queue_insert`

```
TRIGGER: AFTER INSERT on integration_queue

CONDITION: NEW.status = 'pending'

ACTION: Start background task "process_integration_event"

  INPUT: NEW.id
```

Xano Background Task: `process_integration_event`

```
INPUT: queue_id (int)


1. FETCH queue row by id
2. SET status = "processing"


3. SWITCH on destination:


   CASE "sap":
     – Map customer fields → SAP Business Partner schema
     – Map order fields → SAP Sales Order schema
     – Validate required SAP fields (bukrs, vkorg, vtweg)
     – Format as OData JSON or IDoc XML


   CASE "nielseniq":
     – Filter: check consent_records for analytics_partners scope
     – Hash email: SHA-256(lowercase(email))
     – Map Shopify SKU → UPC via product_upc_mapping table
     – Format as NielsenIQ CSV row


   CASE "circana":
     – Same consent + hash as nielseniq
     – Format as Circana JSON schema


   CASE "erp_generic":
     – Apply tenant-specific field mapping from integration_mappings table
     – Validate against destination schema


4. SET status = "ready", transformed_payload = result
```

```
5. POST webhook to Worker:
   POST {worker_url}/events/deliver
   Authorization: Bearer {WORKER_WEBHOOK_SECRET}
   Body: {
     "queue_id": queue_id,
     "event_id": original_event_id,
     "channel": destination,
     "transformed_payload": result,
     "delivery_config": { method, endpoint, credentials_key }
   }

6. On webhook 2xx → SET status = "delivered"
   On webhook fail → INCREMENT retry_count
     IF retry_count >= max_retries → SET status = "dead_letter"
     ELSE → SET status = "pending" (re-triggers task)
```

---

6. Worker: Delivery Endpoint

`POST /events/deliver`

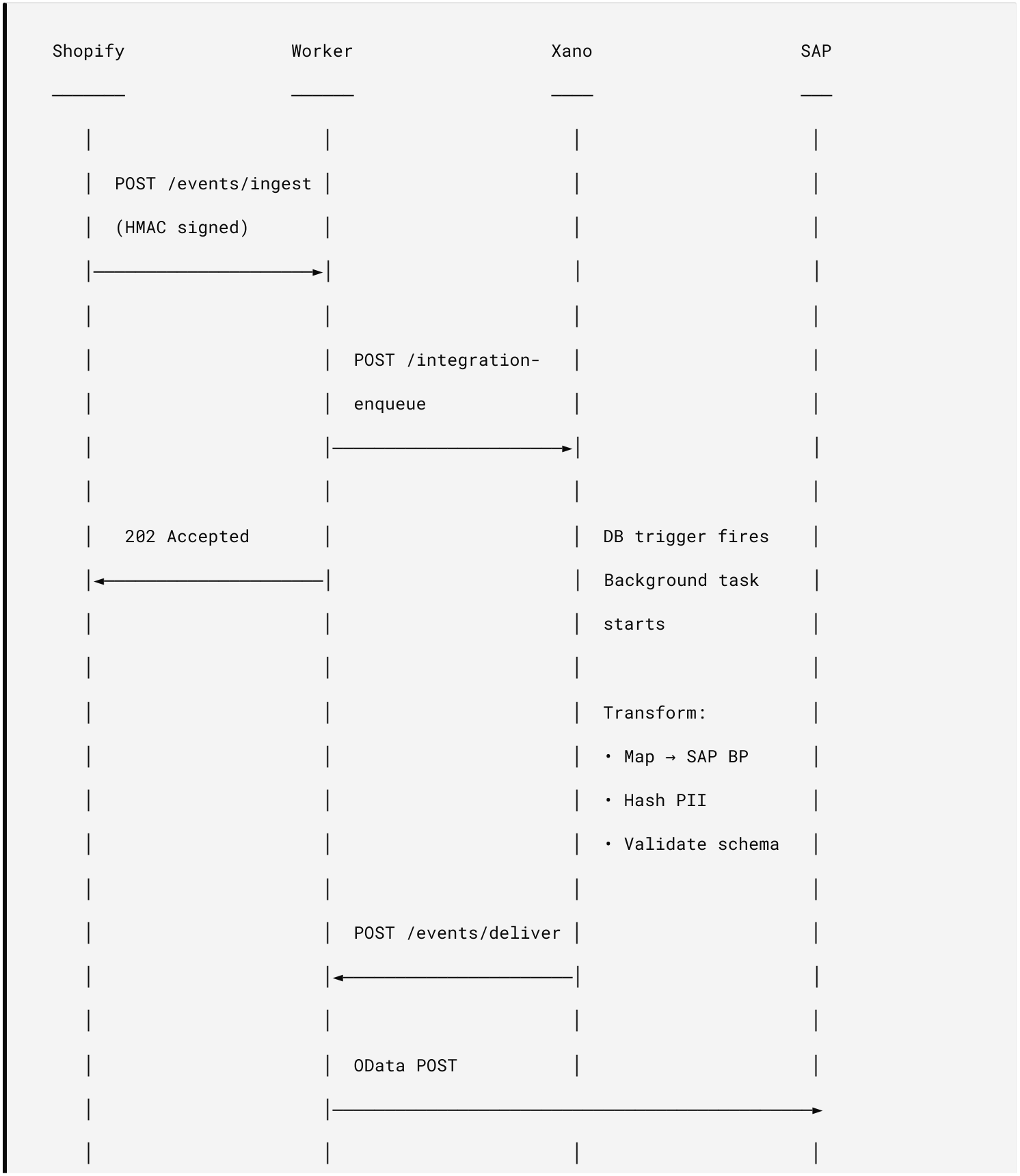
Called by Xano when a message is processed and ready for external delivery:

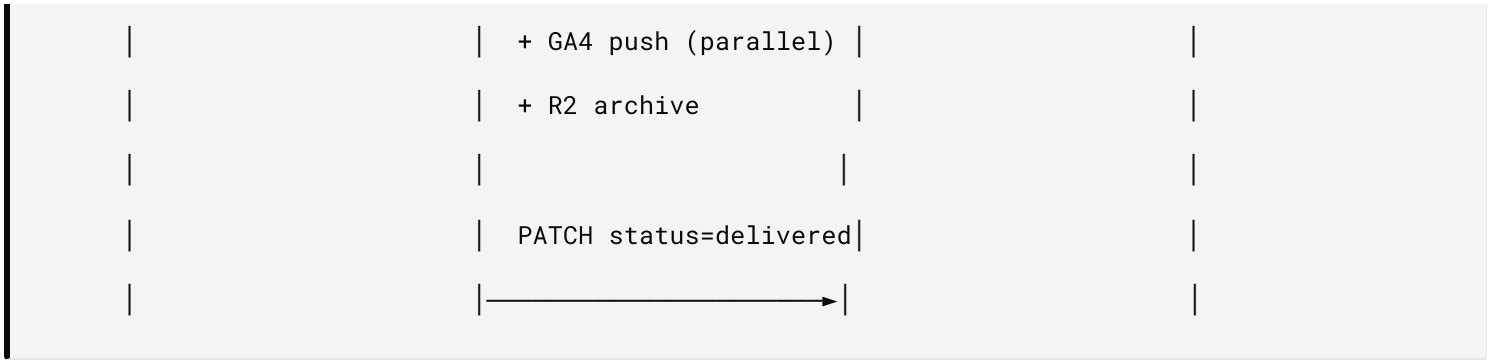
```
async function handleEventDeliver(
  cfg: ResolvedConfig,
  env: Env,
  request: Request,
  origin: string | null
): Promise<Response> {

  // Auth: Xano webhook secret
  const secret = request.headers.get("x-xano-webhook-secret") || "";
  if (secret !== cfg.xanoWebhookSecret) {
    return jsonResponse({ error: "Unauthorized" }, 401, origin);
  }

  const msg = await request.json() as DeliveryMessage;

  // Always archive to R2 (parallel, non-blocking)
  const archivePromise = env.ANALYTICS_EXPORTS.put(
    `events/${msg.channel}/${new Date().toISOString().slice(0,10)}/${msg.event_id}.json`,
    JSON.stringify(msg),
  );

  // Route to delivery handler
  let result: DeliveryResult;
  switch (msg.channel) {
    case "sap":
      result = await deliverToSAP(cfg, env, msg);
      break;
    case "nielseniq":
      result = await deliverToAnalyticsPlatform(cfg, env, msg, "sftp");
```

```
      break;
    case "circana":
      result = await deliverToAnalyticsPlatform(cfg, env, msg, "api");
      break;
    case "ga4":
      result = await deliverToGA4Direct(cfg, msg);
      break;
    case "adobe":
      result = await deliverToAdobeAEP(cfg, env, msg);
      break;
    case "webflow":
      result = await deliverToWebflowCMS(cfg, msg);
      break;
    default:
      result = { ok: true, log: { archived: true } };
  }

  await archivePromise; // ensure archive completes

  return jsonResponse({
    ok: result.ok,
    event_id: msg.event_id,
    channel: msg.channel,
    delivery_log: result.log,
  }, result.ok ? 200 : 502, origin);
}
```

7. Event Flow Examples

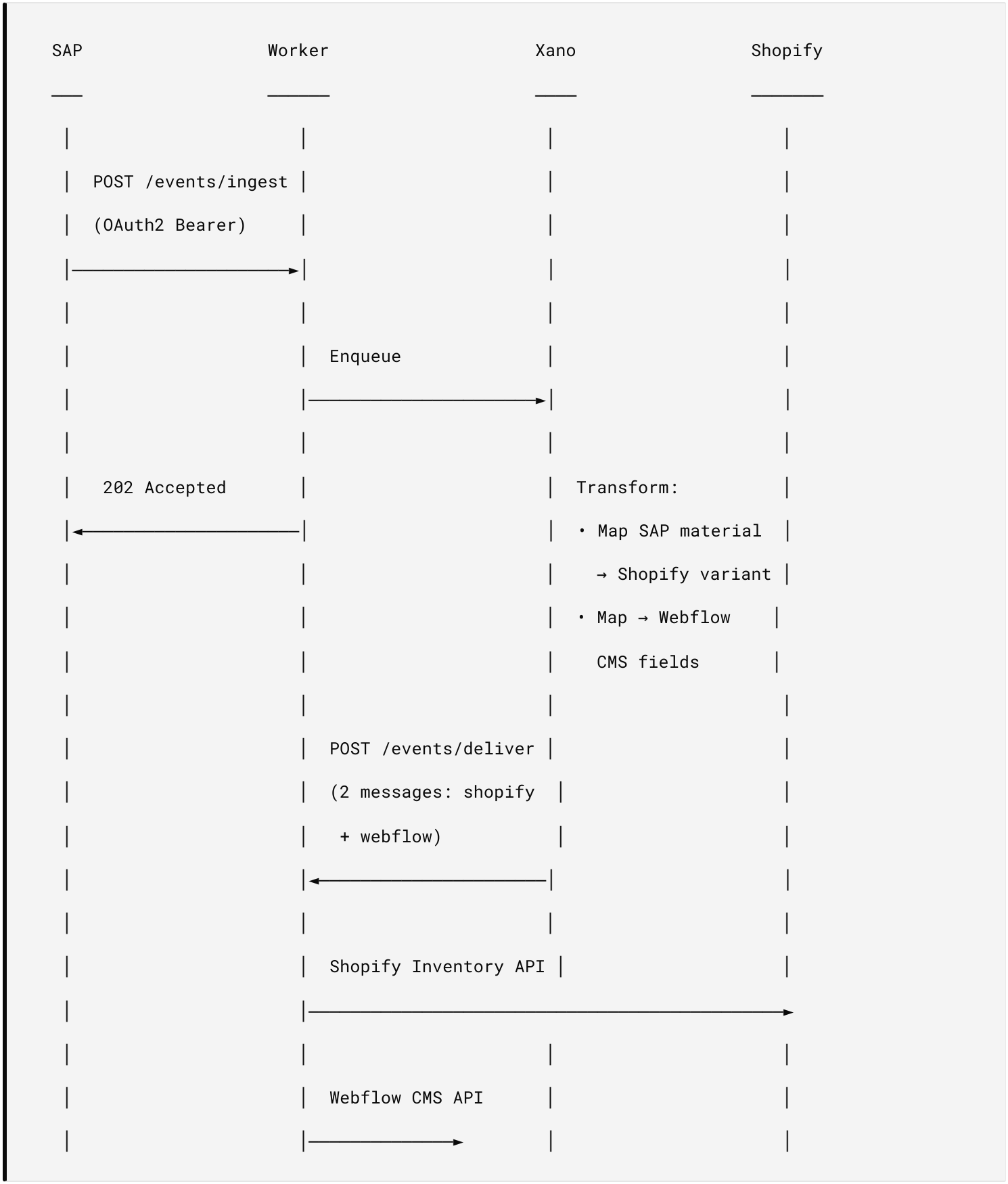
7.1 Shopify Customer Updated → SAP + GA4 + R2



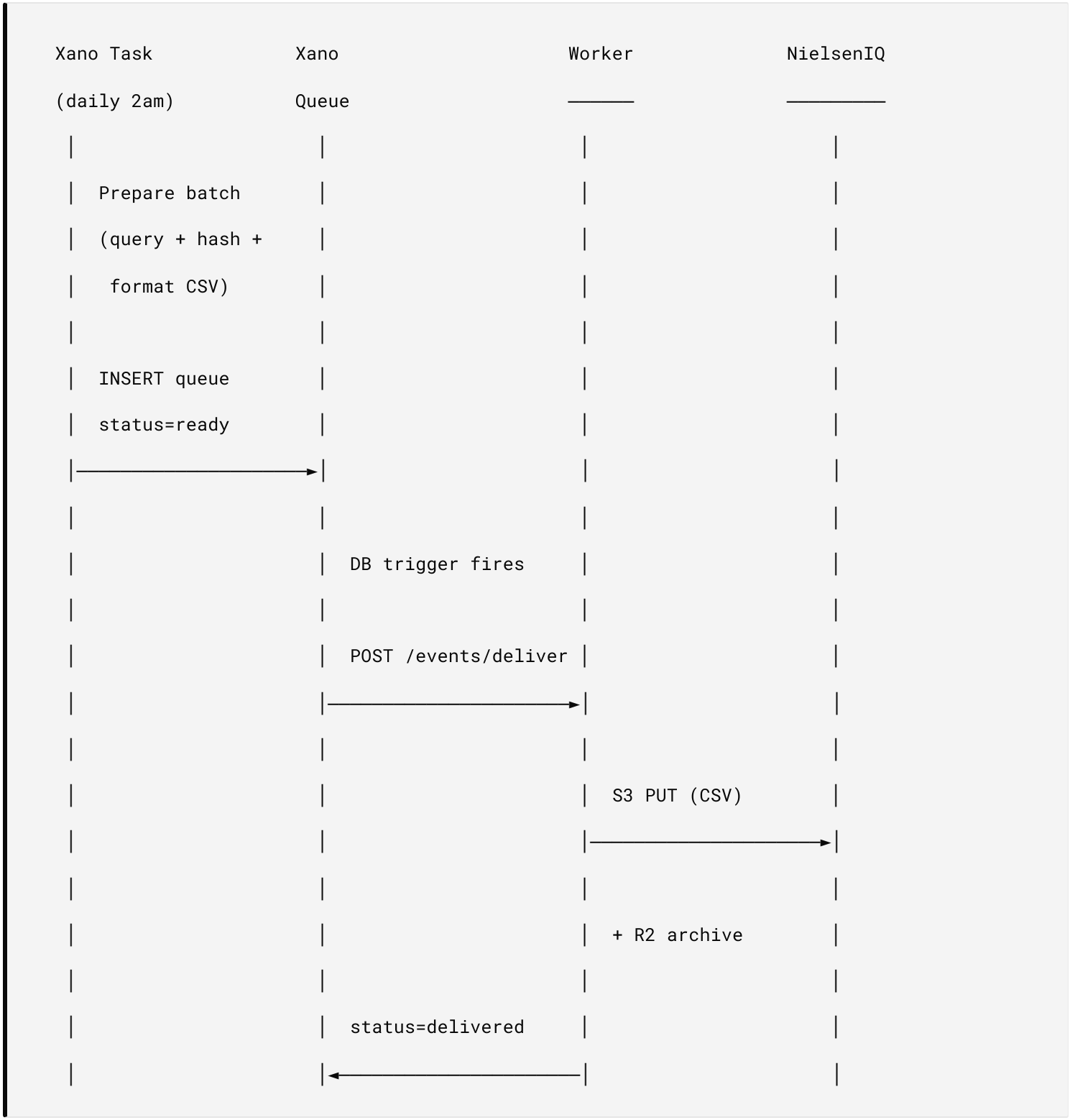


Total latency: 2-5 seconds (vs 15 min with cron polling)

7.2 SAP Inventory Changed → Shopify + Webflow



7.3 Xano Export Ready → NielsenIQ (Batch)



8. What Cron Still Does (Reduced Role)

Cron doesn't disappear — it becomes the **safety net**, not the primary driver:

TASK	BEFORE	AFTER
Customer sync	Cron every 15 min	**Webhook-driven** + cron catches missed webhooks
Token refresh	Cron every 15 min	**Cron** (no event trigger — time-based by nature)
Analytics export	Cron polls Xano	**Xano trigger → webhook** to Worker
SAP data sync	N/A (new)	**Event-driven** only
Dead letter retry	N/A (new)	**Cron** sweep for stuck messages
Stale event cleanup	N/A (new)	**Cron** expire old idempotency keys

```
// Reduced cron handler
async scheduled(_controller: ScheduledController, env: Env, _ctx: ExecutionContext): Promise<vo
  const tenants = await listTenants(env);

  for (const shop of tenants) {
    const cfg = await resolveConfig(env, shop);

    // Token refresh — still time-based (check expiry, refresh if needed)
    await refreshShopifyTokenIfNeeded(cfg, env, shop);

    // Catch-up sync — only processes customers modified since last sync
    // that webhooks may have missed (network failures, etc.)
    await syncCustomersIncremental(cfg, env);

    // Dead letter retry — re-queue failed messages under retry limit
    await retryDeadLetterMessages(cfg, env, shop);
  }

  // Housekeeping — runs once, not per-tenant
  await cleanupExpiredIdempotencyKeys(env);
}
```

9. Failure & Recovery

Retry strategy (per priority)

PRIORITY	MAX RETRIES	BACKOFF	DEAD LETTER AFTER
1 (critical: inventory)	5	10s, 30s, 2m, 10m, 30m	30 min
2 (high: orders)	4	30s, 2m, 10m, 1h	1 hour
3 (normal: customers)	3	2m, 15m, 1h	1 hour
4 (low: tags/segments)	3	15m, 1h, 6h	6 hours
5 (batch: analytics)	2	1h, 6h	6 hours

Dead letter queue handling

```
Dead letter messages are NOT discarded.


1. Stored in Xano: integration_queue WHERE status = 'dead_letter'
2. Archived in R2:  events/dead_letter/{date}/{event_id}.json
3. Alert: POST to admin notification endpoint (email via Resend)
4. Manual replay: POST /admin/events/replay { event_ids: [...] }
5. Cron sweep: retry dead letters older than 1hr with exponential backoff
```

Circuit breaker (per destination)

```
// If a destination fails 5x in 10 minutes, stop sending for 5 minutes
interface CircuitState {
  failures: number;
  last_failure: string;
  open_until: string | null;  // null = circuit closed (healthy)
}

// Stored in KV: circuit:{channel} → CircuitState
// Checked before every delivery attempt
// Auto-resets after cooldown period
```

---

10. Observability

`GET /admin/events/status`

```
{
  "channels": {
    "sap": { "circuit": "closed", "last_24h": { "delivered": 142, "failed": 3, "pending": 0 } }
    "nielseniq": { "circuit": "closed", "last_24h": { "delivered": 1, "failed": 0, "pending": 0
    "ga4": { "circuit": "closed", "last_24h": { "delivered": 847, "failed": 0, "pending": 2 } }
  },
  "dead_letter": { "count": 3, "oldest": "2026-05-27T02:15:00Z" },
  "throughput": { "events_per_minute": 2.3, "avg_latency_ms": 1840 }
}
```

`GET /admin/events/log?channel=sap&limit=20`

Returns recent delivery attempts with status, latency, and error details.

---

11. Migration Path

Phase 1: Add ingest + deliver endpoints (non-breaking)

- New routes: `POST /events/ingest`, `POST /events/deliver`
- Existing webhook handlers unchanged
- Both patterns run in parallel

Phase 2: Redirect Shopify webhooks to unified ingest

- Update Shopify webhook addresses from `/webhooks/customer-update` to `/events/ingest?source=shopify`
- Old endpoints stay as aliases (backward compatible)

Phase 3: Add Xano task runner + DB triggers

- Create `integration_queue` table
- Create background task `process_integration_event`
- Create DB trigger on insert
- Test with R2-only delivery (archive everything, deliver nothing)

Phase 4: Enable external delivery channels

- SAP OData connector
- NielsenIQ S3 drop
- Circana API

- Each channel enabled per-tenant via config flag

Phase 5: Reduce cron to safety net

- Remove analytics polling from cron
- Remove primary customer sync from cron (webhook-driven)
- Keep: token refresh, dead letter sweep, catch-up sync

---

12. Stakeholder Access

CAPABILITY	A (CREATOR)	B (SHARED)	C (PRIVATE)
`/events/ingest`	✓ All sources	✓ Shopify webhooks only	✓ All sources
`/events/deliver`	✓ Xano callback	× Internal only	✓ Their Xano callback
`/admin/events/status`	✓	×	✓
`/admin/events/replay`	✓	×	✓
Configure destinations	✓ Platform config	×	✓ Own config
SAP / ERP credentials	✓ Secrets	×	✓ Own secrets
Dead letter alerts	✓ Email	×	✓ Own alerts

---

13. WMS Socket (v1.1 — 2026-07-22)

The tenant config has carried the socket since the config-connect release — `wms_system` (default `none`) and `wms_base_url` — with no adapter behind it. This

section wires the socket into the bus as a contract, so the first WMS tenant is a config change plus credentials, not a design exercise.

13.1 Transport

WMS platforms deliver into the standard ingest path — `POST /events/ingest` (§4), authenticated per §4 (per-source shared secret or the rotatable key), with `source: "wms"`. No WMS-specific endpoint: the socket IS the envelope.

13.2 Event types

EVENT_TYPE	FIRES WHEN	ENTITY_ID
`unit.received`	Serialized unit checked into a location	serial
`unit.picked`	Unit picked against an order	serial
`unit.packed`	Unit sealed to a shipment	serial
`unit.shipped`	Carrier handoff	serial
`unit.returned`	RMA receipt back into custody	serial

`entity_type: "inventory"` · `priority: 2` — custody events outrank analytics and follow payments.

13.3 Payload contract — references only (the custody rule)

The WMS never holds firmware, keys, or certificates; the bus never accepts WMS PII. A `wms` payload is id/ref-only:

```
{
  "mpn": "HX-DEV-0042",
  "gtin": "00812345678905",
  "sku": "DEV-0042-BLK",
  "serial": "SN-2026-000731",
  "location_code": "US-EAST-1/A-14-3",
  "movement": "inbound",
  "hs_code": "8471.30",
  "country_of_origin": "KR",
  "customs_ref": "MRN-26DE520123456789A1",
  "order_ref": "SHOP-1042",
  "rma_ref": "RMA-2026-0187",
  "carrier_ref": "1Z999AA10123456784"
}
```

`rma_ref` is REQUIRED on `unit.returned` (it joins the custody receipt to the return case) and omitted on all other event types.

`movement` values: `inbound` | `outbound` | `transfer` | `export-cleared` | `import-cleared` | `return` — customs clearance is a movement state on the same five events, not a new event type; a border never reshapes the envelope.

Cross-border ownership boundary: `hs_code` and `country_of_origin` are PIM-owned attributes (one classification per MPN, canonical in the MarketOfSale record) carried on custody events by reference; `customs_ref` (e.g. an MRN) is the declaration pointer. The WMS layer fully owns the movement/custody slice of cross-border data — per-market commercial attributes (price, currency, `eligibleRegion`) stay on the PIM plane, and jurisdiction evidence selection (SBOM + DoC bundles per destination market) stays on the compliance projection; both join the custody stream on `(mpn, serial)`.

No names, no addresses, no contact fields — the WMS keeps those; the bus records custody, not people (the `reconciliation_log` posture). Because the payload is non-personal operational data, WMS events are not consent-gated — they bypass the Consent Mode v2 gate and never fan out to marketing destinations.

13.4 Joins — physical custody ⋈ digital custody

Join keys are `(mpn, serial)`:

- `mpn` → the PIM record (MarketOfSale registry) that owns GTIN/SKU and both catalog projections (channel feed + nested JSON-LD).
- On `unit.shipped` for a firmware-bearing MPN, the bus stamps a ledger row binding `serial → firmware version + sha256` from the vault registry (`firmware_artifacts`) — the CRA Article-14 evidence join: which image was current when this unit left custody.
- `rma_ref` on `unit.returned` -> the return case at `/commerce/returns/{rma}` — customer-service channels answer "the warehouse received your return" from the custody event itself, not from carrier scans; for digital goods the same refund closes the loop by revoking the grant instead.
- A unit's full story = the WMS custody stream ⋈ the firmware unwrap ledger, joined on `(mpn, serial)`.

13.5 Destinations & evidence

`destinations: ["xano"]`, plus BigQuery via the Xano→BQ path. WMS events land in the GENERAL event ledger — append-only by policy until the Sprint-2 hash-chaining (SECURITY-REMEDIATION-PLAN Q2) ships. Until then, custody claims cite the firmware ledger (chained) as primary evidence and present WMS rows as supporting records, not tamper-evident ones.

13.6 Status

SOCKET — config fields live, adapter unwired, no WMS tenant connected. This section is the contract an integrator implements; nothing here ships code.
